Henry Tran

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Education

CUNY City College of New York *New York City, NY*
Bachelor of Science in Computer Science *June 2028*
Coursework: Data Structures, Algorithms, Software Engineering, Linear Algebra, Probability & Statistics

Professional Experience

Kapitus *New York City, NY*
Financial Data Engineer Intern *January - May 2026*

- Re-engineered a daily Alteryx ETL pipeline for tracking loans pledged to financing facilities, consolidating 8 facility-specific macros into one and reducing runtime by 75%
- Built a MySQL view to standardize loan data across financing facilities for portfolio reporting
- Automated freshness and quality checks that caught stale source data before it reached investors
- Documented the redesigned pipeline, successfully handing it off to the team's director and developer

MIT Museum Maker Hub *Cambridge, MA*
Maker & Facilitator *November 2024*

- Designed and built a paper-circuit combination lock with copper-tape sliding switches that light an LED only when the correct code is set; museum staff selected it as a display model for visiting youth
- Mentored K-12 students in drop-in STEAM activities on basic electronics, troubleshooting, and design

Projects

Echo | Python, FastAPI, PostgreSQL, SQLAlchemy, SciPy *October 2026*

- Built and deployed a full-stack web app (echo.henrytran.dev) that matches Spotify listeners who share core taste and can introduce each other to new artists
- Engineered an explainable recommender using IDF-weighted cosine similarity and co-listening scores on sparse user-artist matrices, with two-stage retrieval tested on 150 synthetic listeners
- Designed an idempotent ETL pipeline from the Spotify API to PostgreSQL with OAuth, async fetching, exponential-backoff retries, and artist entity resolution

Leadership & Activities

Tutoring *August 2023 - May 2024*

- **AP Tutor** - Tutored 6 students in AP CS, Calculus, Chemistry, and Statistics; all 6 scored a 5
- **SHSAT Tutor** - Prepared 4 students for the SHSAT; all 4 were admitted to specialized high schools, including The Bronx High School of Science

Robotics Team *August - December 2024*

- **Micromouse** - Wrote C++ firmware implementing flood-fill pathfinding and PID motor control for an autonomous robot that successfully solved a maze

Skills

Languages: Python, SQL, C++, Java, JavaScript, R
Frameworks & Libraries: FastAPI, SQLAlchemy, NumPy, SciPy
Databases & Tools: PostgreSQL, MySQL, Alteryx, Git
Spoken Languages: English, Vietnamese
Interests: Astronomy | Photography | Hiking | Music | Art | Gaming